Project Interim Report

On

File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming With Java

SESSION: 2025-26

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Abstract

- The School Administration Management System is a Java-based project developed using Object-Oriented Programming (OOP) concepts such as inheritance and polymorphism. The main purpose of this project is to manage different roles in a school environment, such as teachers and students, in a structured and efficients way.
- In real-world educational institutions, managing data manually can be time-consuming and error-prone. Therefore, this system is designed to simplify the process by using programming concepts to represent real-life entities. It improves efficiency an reduces complexity by organizing information logically.
- The system uses a base class called person, which contains common attributes like name and age. this class isextended into subclasses techer and student, each having specific properties and behaviors. Method overriding is used to define role-specific actions, and polymorphism is applied to demonstrate dynamic method execution at runtime

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Introduction

Same information about your project but in detail you can also add some of the features of your project. The school administration management system is a java-Oriented Programming conceps such as inheritance and polymorphism. It is designed to manage teacher and student recoard in a simple and efficients way.
This system helps to reduce manual work and provides a structured way to handle information. It into Techer and student classes to demonstrate real-world implementation.

Objectives

- Manage and store teacher and student records effiiently
- Display and track role-based information dynamically
- Reduce manual work and improve accuracy
- Apply Object-Oriented Programming (OOP) concepts such as inheritance and polymorphism
- Improve code resuability and maintainability

Scope:

This project is suitable for small school system to manage techear and student records in a simple and efficients way. It helps in understanding the practical use of object-oriented Programming concepts such as inheritance and Polymorphism.
The system can be further extended by adding features like database integration, graphical user interface (GUI), and additional roles such as admin. It can also be improved to handle larger data and provide move advanced functionalities.

System Requirements:

Hardware: Computer or laptop with minimum 4G RAM, keyboard, and monitor
Software: Java Development Kit (JDK), any IDE such as VS code or Eclipse, Windows Operating system

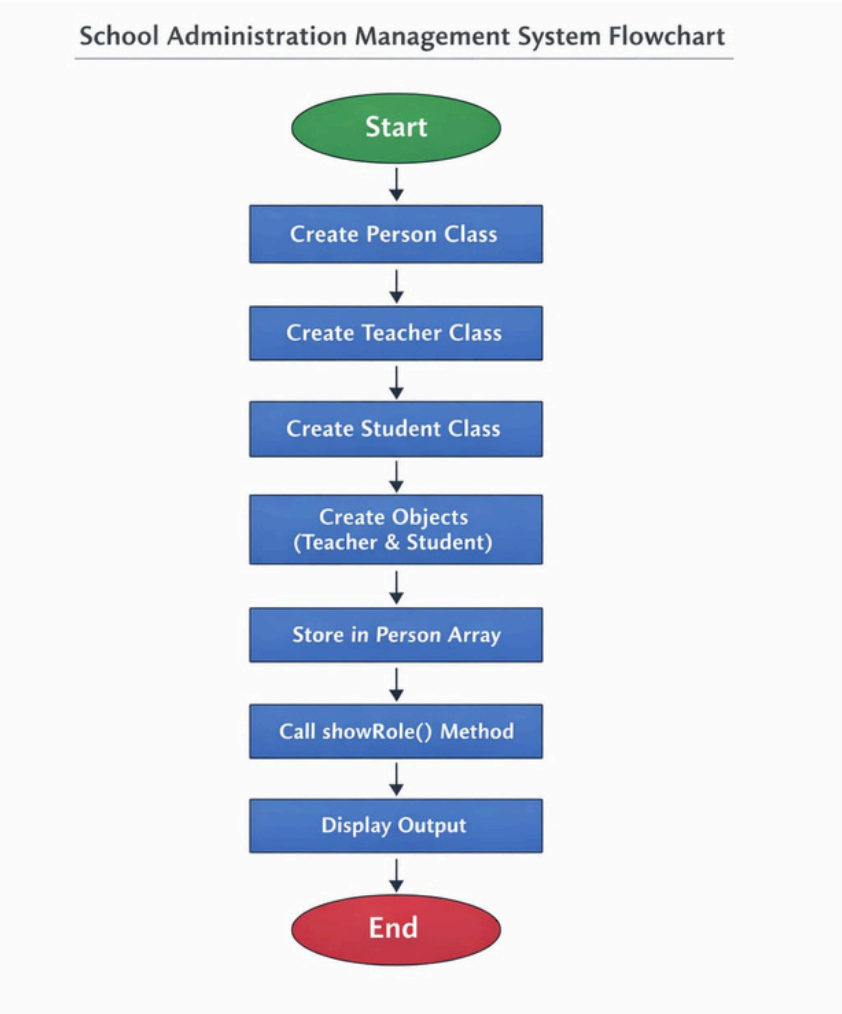
Methodology:

The project works step-by-step to manage teacher and student rols using Object-Oriented Programming Concepts.

1. First, a base class person is created with common attributes like name and age.
2. Then, two subclasses Techear and student are created by extending the person class.
3. Each subclass has its own properties and overrides the method to define specific behavior.
4. Objects of Techears are stored in a person array to demonstrate polymorphism.
5. A loop i used to call the method, and the correct output is displayed based on the object type.

Flowchart:

Includes flowchart (Start, Menu, Choice, Operations), and class Diagram representing system structure. Sample class diagram -



Person Class

. Veriables: Nmae, age

. Methods: ShowRole()

Teacher Class

Variables: subject, salary

.Methods: ShowRole()

Student Class

. Variables: rollNumber, course

. Methods: showRole()

Implementation:

The project is implemented using three main classes:

Person Class: Stores common details such as name and age. It acts as a base class and includes a method showRole() to display a general message.

Teacher Class: Extends the person class and includes additional attributes like subject and salary. It overrides the showRole() method to display teacher-related information.

Student Class: Extends the Person class and includes attributs like roll number and course. It overrides the showRole() method to display student-related information.

The system uses an array of person objects to store Teacher and Student objects. It includes:

.ShowRole() - Displays role-based information
.Loop traversl() - calls methods dynamically for each object
.The program demonstrates runtime polymorphism by calling the same method for different objects and displaying different outputs

.Sample input / Outpute
Output:

1. I am a Teacher. I teach students.
2. I am a student. I study subjects.

Future Enhancements
. Add database for storing records
. Use GUI for better user interface
. Add more roles like admin
. Improve system for handlling large data

Gantt Chart:

Task	Duration
Planning (Understanding OOP Concepts)	1 Day
Design Classes (Person, Teacher, Student)	1 Day
Coding (Inheritance & Polymorphism Implementation)	2 Day
Testing (Checking Output & Logic)	1 Day
Documentation (Report Preparation)	1 Day

1. **planning & Structure**

The project is divided into different tasks such as understanding OOP concepts, designing classes, coding, testing, and documentation. This helps in organizing the developments process in a proper sequence

2. **Time Management**

The Gantt chart shows the time required for each task. It helps in completing the project within the given time and avoids delays.

Conclusion:

The School Administration Management System Successfully demonstrates the use o object-Oriented Programming concepts such As inheritance and polymorphism. The project helps in understanding how different classes like person, Teacher, and student acn be used to represent real-word entities.
It shows how method overrding and runtime polymorphism work in a simple Java program. Overall, the project provides a clear idea of code reusability, flexibility, and real-word pplication of OOP concepts.

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